The Siege
Problem ID: thesiege

You are Warchief Grukk, and today is a good day for war. Your Horde stands ready to crush the Dwarven Underkingdom, N strongholds carved deep into the mountains, each sitting on rich veins of gold.

Your goblin scouts (small, sneaky, and not terribly bright) have somehow wormed their way into every stronghold through cracks and drain pipes the dwarves forgot to seal. They came back clutching crumpled maps scratched onto hide scraps, giggling about "shiny rocks." The maps are crude but surprisingly accurate: each one shows the layout of tunnels inside a stronghold.

Deep inside each stronghold sits the *Gold Vein* (room 0), where the richest ore waits to be mined. Once you conquer a stronghold, your goblin miners will dig out the gold and load it into mine carts that run through the tunnel network to the stronghold entrance (room $V_i - 1$), where your wolf-riders wait with wagons.

Each tunnel has a *weight limit*: the maximum kilograms of gold ore that a mine cart can carry through it per hour. Some tunnels are wide enough for a troll to stroll through; others are barely goblin-sized. Your miners will haul as much gold as physically possible from the vein to the entrance, using every tunnel to its limit simultaneously. You need to figure out how much gold each stronghold can yield.

But strongholds do not fall for free. The dwarves fight like cornered badgers behind their stone barricades. Each assault costs orc warriors their lives. You can muster exactly B warriors for the entire campaign, and you must conquer your chosen strongholds before the elven relief army arrives. Choose which strongholds to take to maximise the total gold your miners extract. Each stronghold can only be taken once.

Present your *Siege Plans* to the war council: the list of strongholds the Horde will conquer.

Input

The first line contains two integers N and B ($1 \leq N \leq 50$, $1 \leq B \leq 1000$), the number of dwarven strongholds and the number of orc warriors available.

Then N stronghold descriptions follow. Each stronghold starts with a line containing:

- a string s_i, the name of the stronghold (lowercase `a`–`z`, no whitespace, length ≤ 20),
- an integer c_i ($1 \leq c_i \leq B$), the number of orc warriors lost conquering it,
- an integer V_i ($2 \leq V_i \leq 15$), the number of rooms,
- an integer E_i ($1 \leq E_i \leq 40$), the number of tunnels.

The next E_i lines each describe one tunnel with three integers u, v, w ($0 \leq u, v < V_i$, $u \neq v$, $1 \leq w \leq 100$), meaning a one-way tunnel from room u to room v with weight limit w.

Room 0 is the Gold Vein (source) and room $V_i - 1$ is the entrance (sink).

Output

Output the strongholds you should conquer to maximise total gold extracted without exceeding the warrior budget B. Print one line per selected stronghold in the format:

```
<index> <name>
```

where `<index>` is the 0-based position of the stronghold in the input. Lines must appear in ascending order of index. If multiple subsets achieve the maximum gold, output the one with the lexicographically smallest sequence of indices.

Sample Input 1

```
3 10
ironkeep 3 3 2
0 1 5
1 2 2
stonewall 6 4 4
0 1 3
0 2 2
1 3 2
2 3 3
thornhold 4 3 2
0 1 4
1 2 3
```

Sample Output 1

```
1 stonewall
2 thornhold
```

Sample Input 2

```
3 10
dragonspire 6 4 4
0 1 4
0 2 2
1 3 4
2 3 3
ravencrest 4 3 3
0 1 5
0 2 3
1 2 2
wolfgate 4 3 2
0 1 7
1 2 5
```

Sample Output 2

```
0 dragonspire
1 ravencrest
```

Sample Input 3

```
1 5
lastbastion 3 3 2
0 1 4
1 2 4
```

Sample Output 3

```
0 lastbastion
```